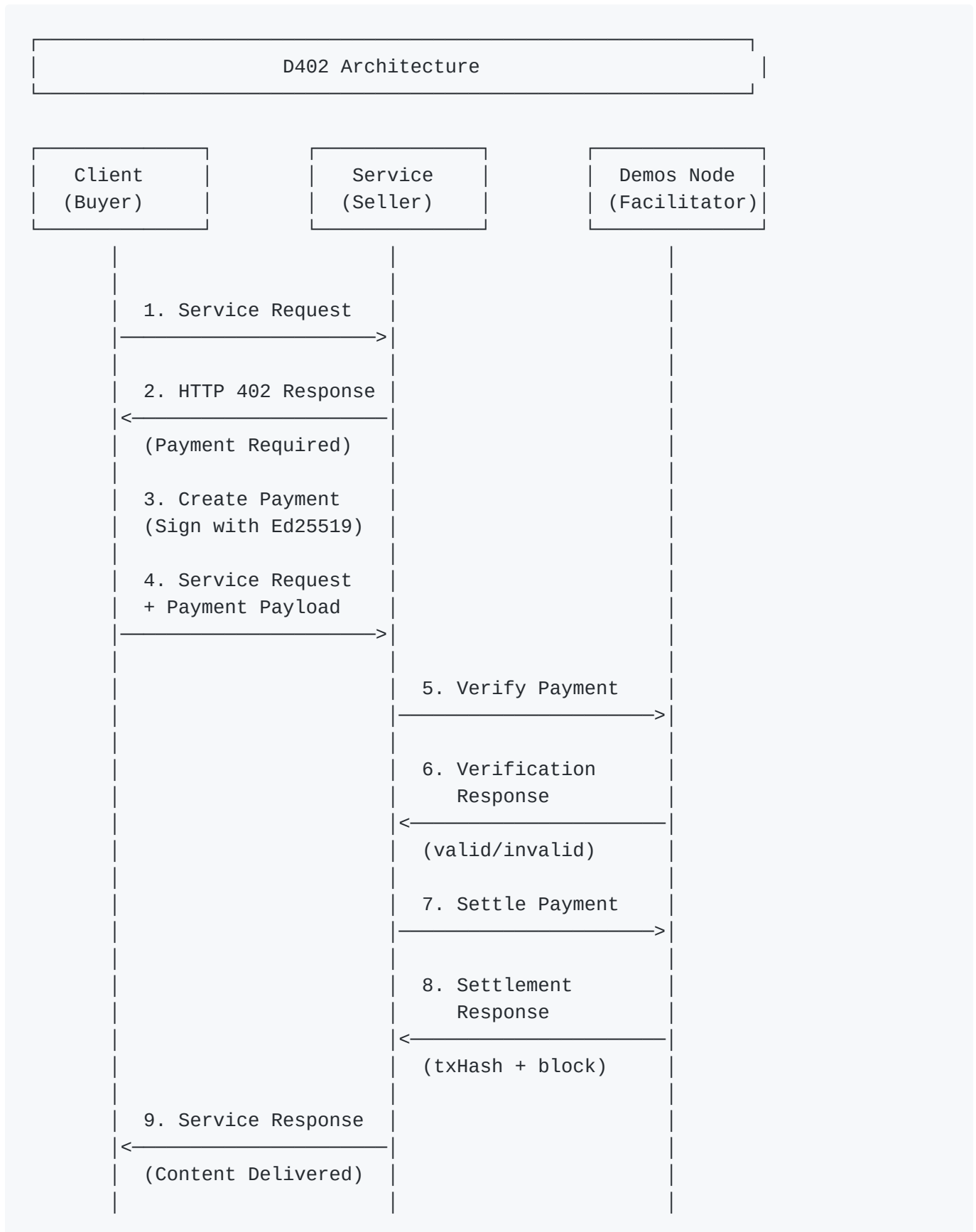


x402d

System Components



Component Responsibilities

Client (Buyer):

- Creates payment transactions
- Signs transactions with Ed25519 private key
- Includes payment proof in service requests
- Uses Demos SDK D402 client library

Service (Seller):

- Returns HTTP 402 with payment requirements
- Validates payment proofs via facilitator
- Requests settlement after successful validation
- Delivers content upon settlement confirmation

Facilitator (Demos Node):

- Verifies transaction signatures
- Checks sender balances in GCR
- Validates transaction structure and nonces
- Executes settlement via Transaction objects in the Mempool
- Returns settlement receipts

Network Architecture

- **Primary Network:** Demos Network mainnet
- **Token:** DEM (native blockchain token)
- **Consensus:** PoR (Proof of Representation) BFT
- **State Management:** GCR (Global Change Registry)
- **Balance Storage:** Blockchain Native GCR tables
- **Finality:** Fast finality